

M.Tech. (Integrated) DEGREE EXAMINATION, DECEMBER 2022
Second and Third Semester

21CSC101T – OBJECT ORIENTED DESIGN AND PROGRAMMING
(For the candidates admitted from the academic year 2021-2022)

Note:

- (i) **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.
- (ii) **Part - B** should be answered in answer booklet.

Time: 3 Hours

Max. Marks: 75

Marks BL CO PO

PART – A (20 × 1 = 20Marks)

Answer ALL Questions

1. What happens when we try to compile class definition in following code?

```
class Birds {};
```

```
class Peacock : protected Birds {};
```

(A) It will not compile because class body of Birds is not defined
 (B) It will not compile because class body of peacock is not defined
 (C) It will not compile because a class cannot be protectedly inherited from other class
 (D) It will compile successfully

1 2 1 2,3
5
2. Which of the following statement is correct regarding destructor of base class?

(A) Destructor of base class should always be static
 (B) Destructor of base class should always be virtual
 (C) Destructor of base class should not be virtual
 (D) Destructor of base class should always be private

1 1 1 2,3
5
3. Which of the following access specifiers is used in a class definition by default?

(A) Protected
 (B) Private
 (C) Public
 (D) Friend

1 1 1 2,3
5
4. Which of the following statements is correct about program given below.

```
class Bix
```

```
{public :
```

```
static void myfunction (); };
```

```
int main ()
```

```
{ void (*ptr) () = & Bix ::Myfunction;
```

```
return 0; }
```

(A) Program reports an error as pointer to member function cannot be defined outside definition of class
 (B) Reports an error as pointer to static member function cannot be defined
 (C) Reports an error as pointer to member function cannot be defined without object.
 (D) The program reports linker error

1 2 1 2,3
5

5. Which among the following can be heuristic for use case diagram?

- (A) Product can be made actor (B) Never name actors with noun phrases
(C) Name use cases with verb (D) Product can be use case phrases

6. An operation can be described as?

- (A) Object behavior (B) Class behavior
(C) Functions (D) Object and class behavior

7. Find the o/p of the following code

```
class cube {public: int side; cube ( ) {side = 10;}};  
int main ( ) {  
    cube c;  
    cout << c.side;}  
}
```

- (A) 11 (B) 2
(C) 3 (D) 10

8. _____ are special type of constructors which takes an object as argument, and its used to copy values of data members of one object into another object

- (A) Default constructor (B) Parameterized constructor
(C) Constructors (D) Copy constructors

9. Which among the following best describes constructor overloading?

- (A) Defining one constructor in each class of a program (B) Defining more than one constructor in single class
(C) Defining more than one constructor in single class with different signature (D) Defining destructor with each constructor

10. Using friend operator function, following perfect set of operators may not be over loaded.

- (A) =, (), [], → (B) <<, ==, [], >>
(C) ?, =, (), ++ (D) +, -, --, ++

11. To model life time of a reactive system _____ UML diagrams can be used.

- (A) Activity (B) Usecase
(C) Class (D) Statechart

12. Which among the following is correct for multiple inheritance?

- (A) class student {public : int marks;} s; class stream {int marks;} stream {}; class topper: public student, public stream {};
(B) class student {int marks;} s; class stream {int marks;} stream {}; class topper: public student {};
(C) class student {int marks;} s; class stream {int marks;} stream {}; class topper: public student {};
(D) class student {} s; class stream {} stream {}; class topper: public student {};

13. Which of the following combines two concurrent activities and re-introduces them to a flow where only 1 activity can be performed at a time? 1 2 3 2,3
.5
- (A) Fork symbol (B) Joint symbol
(C) Decision symbol (D) Note symbol
14. Class X, class Y and class Z are derived from class base. This is inheritance. 1 2 3 2,3
.5
- (A) Multiple (B) Hierarchical
(C) Multilevel (D) Single
15. Predict the output: 1 2 3 2,3
.5
- ```
class base { int arr [10];};
class b1 : public base {};
class b2 : public base {};
class derived : public b1, public b2 {};
main () {cout << size of (derived); return;}
```
- (A) 40 (B) 80  
(C) 0 (D) 4
16. Which of the following best defines the syntax for template function? 1 2 4 2,3  
.5
- (A) template return\_type function\_Name (Parameters)  
(B) template return\_type function\_name (parameters)  
(C) Both A and B (D) template return\_type
17. Predict the output template <int i> 1 2 4 2,3  
.5
- ```
void fun () {i=20; cout <<i;}
int main () {
    fun<10>();
    return 0; }
```
- (A) 10 (B) 20
(C) compile error (D) 5
18. Return type of `uncaught_exception ()` is 1 2 4 2,3
.5
- (A) int (B) bool
(C) char * (D) double
19. Predict the output 1 2 5 2,3
.5
- ```
main () {try {throw 'b' ;}
 catch (int param) {cout<< "int exception";
 }
 catch (...) {cout << "default exception";
 }
 cout << "After Exception"; return o;}
```
- (A) int exception after exception (B) int exception  
(C) default exception (D) default exception after exception
20. What is a collection of model elements called? 1 2 5 2,3  
.5
- (A) Box (B) Dependency  
(C) UML packages (D) Package members

**PART – B (5 × 8 = 40 Marks)**

Answer ALL Questions

| Marks | BL | CO | PO       |
|-------|----|----|----------|
| 8     | 4  | 1  | 2,3<br>5 |

21. a. Construct a C++ program to create a class employee with datamembers as Employee\_name, EID, phone, DoJ, no\_of\_hours-worked, pay\_rate. Write necessary member functions to calculate and display the salary of employee. (Use default value 10,000 for pay-rate). Apply the concept of default, parameterized and copy constructor to develop this complete code in C++.

**(OR)**

- b. Rahul and Guldeep plays a mathematical game between each others. The game is all about finding root of quadratic equations. The concept of constructor need to be used for getting input values.

|   |   |   |          |
|---|---|---|----------|
| 8 | 4 | 1 | 2,3<br>5 |
|---|---|---|----------|

The constraints are

$$1 \leq a \leq 50$$

$$1 \leq b \leq 50$$

$$1 \leq c \leq 50$$

Write a C++ program to get the described output for above scenarios.

22. a.i. Create a class time which contains \_Hours, \_Minutes, \_seconds. Write a C++ program using operator overloading for the following:

|   |   |   |          |
|---|---|---|----------|
| 4 | 4 | 2 | 2,3<br>5 |
|---|---|---|----------|

1) == : To check whether two time are same or not.

2) >>: To accept the time

3) << : To display the time

- ii. Draw sequence diagram for car rental system. Where the admin able to login into the system using their credentials? After login user can manage all the operations passengers, drivers, cars, booking, car routes, payment gateway. Identify various objects involved in this project. Draw sequence and collaboration diagram for the same project. Explain the components in detail.

|   |   |   |          |
|---|---|---|----------|
| 4 | 4 | 2 | 2,3<br>5 |
|---|---|---|----------|

**(OR)**

- b.i. Write a program in C++ to check the given number is Armstrong number or not.

|   |   |   |          |
|---|---|---|----------|
| 4 | 4 | 2 | 2,3<br>5 |
|---|---|---|----------|

A student is given a task to overload == operator.

Armstrong number is a number that is equal to sum of cube of its digits.

Input is an integer ranges  $1 \leq N \leq 8000$ . If the input is Armstrong, print Armstrong number otherwise print not an Armstrong number.

- ii. Draw collaboration diagram for the following scenario.

|   |   |   |          |
|---|---|---|----------|
| 4 | 4 | 2 | 2,3<br>5 |
|---|---|---|----------|

\*students of a university have membership in library.

\*Librarian verifies the membership account.

\*If the account is valid, then student requests a book.

\*The librarian checks the availability of the book.

\*If the book is available, it will be issued to the student.

\*The Librarian processes the fine details. Explain the components used.



23. a.i. Write a program to calculate the total marks of each students of a class in physics, chemistry and maths and average marks of the class. The number of students in the class are entered by the user. Create a class named marks with data members for roll number, name and marks. Create three other classes inheriting the marks class namely phy, che and maths which are used to define marks in individual subject of each student. Implement the above scenario using inheritance concepts.

4 4 3 2,3  
,5

ii. You are required to present online design of a system that will be used by doctors at a medical practice to keep patient records. The system should record each consultation between a doctor and a patient, any illness diagnosed by the doctor, and any drugs prescribed to the patient. At regular intervals, an auditor will use the system to check whether the same drug is bring prescribed repeatedly to a particular patient. Sketch a diagram. Explain the components method.

4 4 3 2,3  
,5

**(OR)**

b.i. Construct a C++ program using class and object to implement Fibonacci series of given number using recursion concept and abstract class concept but the condition is, program has to refer function in cache memory and not from main memory.

4 4 3 2,3  
,5

ii. Construct statechart diagram for accessing payroll details for particular employee.

4 4 3 2,3  
,5

24. a.i. Write a program to demonstrate the use of special functions, constructor and destructor in class template. The program is used to find the biggest of two entered numbers.

4 4 4 2,3  
,5

ii. Supply code for computeAge ( ) function such that an exception of type invalid input exception is thrown whenever either of two inputs is negative or if the current year is less than date of birth.

4 4 4 2,3  
,5

**(OR)**

b.i. Write a program to define function template for calculating square of given numbers with different datatypes.

4 4 4 2,3  
,5

ii. Implement a main function such that, enclosed within a try statement, the computeAge() function is executed with invalid inputs. A catch statement should catch the exception, print out an error message and cause the program to immediately exit.

4 4 4 2,3  
,5

25. a. Write a C++ code to demonstrate sorting of 2D vector on basis of number of columns in ascending order.

8 4 5 2,3  
,5

**(OR)**

b. Write a C++ code using fstream class object to write the content in file and then to read the content of file.

8 4 5 2,3  
,5

**PART – C (1 × 15 = 15 Marks)**

Answer **ANY ONE** Questions

Marks BL CO PO

26. Consider an electricity bill statement (EBS) real time application. The EBS calculates electricity charger (EC) from the units consumed by consumer. Construct class and usecase diagram for above diagram.

15 4 1 2.5,  
5

27. Create usecase and class diagram for library management system.

15 4 1 1.2,  
3.5

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